Idriss Khattabi

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Location: Available across Morocco



Education

Master in Artificial Intelligence and Data Science — Oct. 2023 – July 2025
Faculty of Sciences and Technologies — Tangier, Morocco

Bachelor in Computer Science and Mathematics — Oct. 2019 – July 2023
Faculty of Poly-disciplinary — Larache, Morocco

Experience

Data / Digitalization Engineer Intern, KOSTAL Group – Tangier — Jan. 2026 – Present

- Analyzed production data to identify bottlenecks, defect patterns, and performance drifts, while exploring AI-driven approaches for anomaly detection, predictive insights, and intelligent process monitoring.
- Built robust Data Pipelines to extract, clean, transform, and consolidate noisy data from multiple operational sources into structured datasets.
- Developed a digital platform for Project Launch Management, improving project tracking, reporting, workflow visibility, and collaboration across teams, and developed an AI agent to analyze project and operational data for actionable insights.

AI & Data Engineer Intern, Bid2Win – Remote, Part-Time — Dec. 2025 – May 2026

- Developed an AI-powered platform to automate tender analysis, improve document filtering, and support scoring and recommendation workflows.
- Built automated Data Pipelines for scraping, ingesting, cleaning, transforming, chunking, and indexing unstructured tender documents from multiple sources.
- Implemented a RAG-based feature-extraction pipeline to identify tender requirements, eligibility criteria, and relevant insights for scoring and recommendation.

AI & Data Engineer Intern, High Commission for Planning – Remote — Sept. 2025 – Dec. 2025

- Designed and developed an AI-powered platform to automate the documentation and management of statistical processes based on the **GSBPM** framework.
- Engineered document-ingestion and data-processing pipelines to collect, process, structure, and index documents from multiple sources.
- Implemented a RAG engine using **HyDE** prompting and semantic retrieval to generate structured, GSBPM-compliant documentation.
- Automated info-extraction and report-generation workflows, reducing statistical documentation time.

AI & Data Engineer Intern, TNA Consulting – Casablanca — Feb. 2025 – July 2025

- Developed an AI-powered platform enabling financial institutions to search, interpret, and leverage African central-bank publications and regulatory documents.
- Engineered scalable Data Pipelines for scraping, extracting, cleaning, transforming, chunking, embedding, and indexing large volumes of financial documents.
- Built a RAG-based chatbot capable of answering user questions using contextual information retrieved from central-bank publications.

Research Assistant, Faculty of Poly-disciplinary – Larache — Mar. 2023 – June 2023

- Developed an optimization system to improve the consistency of Analytic Hierarchy Process (AHP) pairwise-comparison matrices using Genetic Algorithms (GA) and Particle Swarm Optimization (PSO).
- Built a web application for automated AHP matrix generation, validation, and decision-support workflows.
- Co-authored **three** research papers with **Dr. Chakir Tajani**: Paper 1 – Paper 2 – Paper 3

Languages

Arabic: Native proficiency | **English:** Professional (B2) | **French:** Professional (B2)

Certifications

Associate Data Scientist in Python – DataCamp
Associate AI Engineer for Data Scientist – DataCamp
Microsoft Azure Fundamentals – DataCamp
Associate Data Engineer – DataCamp

Skills

- **Programming Languages:** Python, SQL, Java, JavaScript, C/C++, Bash
- **Big Data & Streaming:** Apache PySpark, Apache Kafka, Apache Airflow, RabbitMQ
- **Machine Learning & Generative AI:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, MLflow, LangChain, LangGraph, CrewAI, Agno, RAG, Prompt Engineering, AI Agents, Ollama, OpenAI API, Fine-Tuning
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Snowflake, ChromaDB, Qdrant, Weaviate
- **Backend & APIs:** FastAPI, Django, Flask, Streamlit, REST APIs, Selenium
- **DevOps & Cloud:** Docker, Docker Compose, Git/GitHub, CI/CD, Linux, Azure, Postman
- **Soft Skills:** Analytical Thinking, Problem Solving, Fast Learning, Teamwork, Communication

Projects

Real-Time Twitter Sentiment Analysis [GitHub Repo]

- Developed a real-time streaming Data Pipeline using Kafka and Spark Streaming to classify tweets.
- Implemented stream-processing workflows for data ingestion, transformation, sentiment classification.
- Built live analytics dashboards to visualize sentiment trends and system performance metrics.
- Tools Used: PySpark, Kafka, Docker, Django, MongoDB, ChartJS.

AI Scraper – AI-Powered Data Extraction Pipeline [GitHub Repo]

- Developed an AI-powered web-scraping application to extract, clean, and structure data from static and dynamic websites using local or cloud-based LLMs.
- Built an automated pipeline to process unstructured web content and transform it into structured datasets for analysis and downstream AI workflows.
- Tools Used: Python, Selenium, Crawl4AI, Streamlit, Ollama, SambaNova API, Pandas.

Data Space for Electronic Medical Records [GitHub Repo]

- Developed a centralized platform for managing, processing, and exploring electronic healthcare data.
- Designed REST APIs and ETL workflows for secure ingestion, transformation, storage, and retrieval of medical records.
- Improved accessibility and interoperability of healthcare data through centralized data features.
- Tools Used: Python, MySQL, REST APIs, ETL Pipelines.

Data Analysis & Machine Learning Tool-Kit Desktop Application [GitHub Repo]

- Developed a Python desktop application that enables users to analyze datasets and apply Machine Learning algorithms through an intuitive graphical interface.
- Implemented data exploration, preprocessing, visualization, and Machine Learning workflows for classification, regression, and clustering tasks.
- Tools Used: Python, Tkinter, Scikit-learn, Pandas, Matplotlib.

ChatPDF – Local RAG Document Assistant [GitHub Repo]

- Developed a local AI assistant that enables users to ask natural-language questions about PDF documents.
- Designed an end-to-end RAG pipeline covering PDF parsing, document chunking, embedding generation, Vector Database retrieval, and LLM-based answer generation.
- Tools Used: Python, Streamlit, LangChain, Ollama, SambaNova API, ChromaDB, Marker.